# Chinnolla Koteshwar

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| **Date of birth** | 21 March 2004 | **Email** | chinnollakoteshwar@gmail.com |
| **Place of birth** | Hyderabad, India | **Phone** | +91 6300727875 |
| **Nationality** | Indian | **LinkedIn** | ln/koteshwar-chinnolla |
| **Gender** | Male | **GitHub** | KoteshwarChinnolla |
| **Marital status** | Single | **Portfolio** | koteshwarchinnolla.github.io |


## Research Profile

Electronics and Communication Engineering graduate with a minor in Artificial Intelligence and Machine Learning, with research experience in autonomous robotics and natural language-driven intelligent systems. My academic work focuses on integrating language models with decision-making systems for real-world robotic applications. Alongside research, I have over one year of experience in backend engineering, cloud infrastructure, and scalable AI system deployment.

## Research Interests

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| **Autonomous Robotics** | **Natural Language Systems** | **Reinforcement Learning** |
| **Intelligent Agents** | **Computer Vision** | **Autonomous Systems** |

## Relevant Coursework

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| **AI & Machine Learning** | Artificial Neural Networks, Machine Learning, Python |
| **Computer Science** | Data Structures, Object-Oriented Programming (Java), Cloud Computing, DBMS |
| **IoT & Embedded Systems** | Internet of Things, Microcontrollers |
| **Mathematics** | Linear Algebra, Differential Calculus, Probability & Stochastic Processes |
| **Control & Signals** | Linear Control Systems, Digital Signal Processing, Signals & Systems |
| **Core Electronics** | Analog & Digital Circuits, VLSI, Networks |

## Education

**B.Tech** – Electronics & Communication Engineering *(Minor: AI/ML)* — **May 2025**
*Gokaraju Rangaraju Institute of Engineering and Technology (GRIET), Hyderabad* — CGPA: **8.39 / 10**

**Senior Secondary (Intermediate)** — **June 2021**
*NSR Impulse Junior College* — 988 / 1000

**Secondary School Certificate** — **March 2019**
*Matrusri High School* — 9.8 / 10

## Publications & Research


[1] **C. Koteshwar**, M. D. Chaithanya, M. Abdul Kaleem, and A. Vishnubhatla, *"Natural Language Based Autonomous Navigation Vehicle,"* in *Proc. 2025 International Conference on Computing Technologies (ICOCT)*, Bengaluru, India, 2025, pp. 1–7.
**DOI:** 10.1109/ICOCT64433.2025.11118685 • **Venue:** IEEE ICOCT 2025 • **Keywords:** *LLMs, Computer Vision, Natural Language, Autonomous Navigation*

## Work Experience

**Associate Software Engineer – AI & DevOps** *Anasol Consultancy Services, Hyderabad* Apr 2025 – Present

- Built and maintained AI-enabled backend systems for enterprise applications.
- Worked on recommendation systems and agent-based workflows for internal automation.
- Developed scalable infrastructure using AWS, Kubernetes, Terraform, Helm, Redis, Kafka, and microservice design patterns.
- Created reusable open-source Terraform modules and Helm charts for production-ready cloud deployments.

## Projects

**ApiSynIQ** Go, Python, Java, gRPC, PostgreSQL (pgvector), LangChain, LangGraph

- Developed a multi-service framework for natural language interaction with REST APIs, that lets users interact with APIs using natural voice or text instead of traditional forms.
- Designed a multi-service architecture to understand user intent, identify the right endpoint, and generate valid API requests automatically.
- Implemented a semantic API discovery engine with pgvector and RAG, enabling the system to match human language to the correct backend service.
- Created a developer-friendly annotation SDK that turns existing APIs into conversational interfaces with minimal code changes.

**Natural Language Based Autonomous Navigation Vehicle** Python, Raspberry Pi, Arduino, OpenCV, A*, Docker, LLMs

- Developed an autonomous vehicle that interprets spoken or text commands into task-specific navigation workflows in indoor environments.
- Integrated LLM-based planning and response modules to interpret user intent, ask clarification questions, and manage multi-step tasks dynamically.
- Implemented A* path planning with a custom motion decoder, enabling accurate obstacle-aware navigation and object delivery between locations.
- Published the work at IEEE ICOCT 2025 for its approach to combining conversational AI with robotic task execution.

**Cloud Infrastructure Automation Toolkit** Terraform, Helm, Kubernetes, AWS, ArgoCD, Istio - Terraform - HELM

- Built reusable Terraform modules and Helm charts to automate cloud infrastructure provisioning and Kubernetes deployments across AWS environments.
- Designed production-ready templates for EKS, ECS, VPC, RDS, ALB, Kafka, Redis, and microservices, reducing repetitive setup for scalable systems.
- Integrated GitOps workflows with ArgoCD and Helm for consistent, version-controlled deployments from infrastructure to application layer.

**NLP Architecture Evolution Study** Python, TensorFlow, PyTorch, Scikit-learn, Transformers - BERT - NLP Techniques - RNNs

- Developed three connected NLP projects covering spam classification, BERT pre-training, and sentiment analysis to understand the evolution from statistical methods to deep contextual learning.
- Implemented and compared Bag of Words, TF-IDF, Word2Vec, Bi-LSTM, and BERT, observing how each architecture improved language understanding and prediction accuracy.
- Built models from the ground up to study internal behavior of each technique rather than treating them as black-box libraries.

**OpenCV Computer Vision Learning Repository** Python, OpenCV, YOLOv8, NumPy

- Built a practical computer vision repository exploring core image processing concepts and real-time object detection workflows.
- Developed a vehicle counting system using line-cross detection and object tracking to monitor cars in video streams.
- Integrated YOLOv8-based object detection on images and videos to study modern deep learning approaches in computer vision.

## Technical Skills

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| **Programming** | Python, C++, Java, Go |
| **AI & Research** | Deep Learning, TensorFlow, PyTorch, OpenCV, keras, NLP, Reinforcement Learning, RAG, LangChain, LangGraph |
| **Robotics** | ROS, Raspberry Pi, Arduino, Computer Vision |
| **Cloud & Infrastructure** | AWS, Kubernetes, Terraform, Helm, Docker, Kafka, Redis |

## Leadership & Activities

- **Mentorship (Advanced Academic Center):** Mentored 4 students in AI/ML, guiding them from initial concepts to successful project implementations and final submissions.
- **Problem Solving:** Solved **480+** problems across Java, Python, C++, and Go on LeetCode.

## Hackathons

- Participated in 4 technical hackathons.
- **Winner:** AAC Product Demand Prediction Challenge.
- **CBIT SUDHEE 23:** ML-integrated healthcare system.
- **EPITOME 23:** MedTech.
- **CHALLENGE ACI:** Blockchain-based Electronic Health Record Management.

## Training

- Agentic AI – Krish Naik Academy.
- AWS Cloud – AWS Skill Builder.
- Machine Learning – Udemy.

## Language Proficiency

**Language Proficiency – English** **2026**
*IELTS* 6.5

**Language Proficiency – Telugu**
*Native* N/A

**Language Proficiency – Hindi**
*Fluent* N/A